

Senior Software Engineer Exercise

File Import and Export

Context

You are building a backend service that receives data files from an external system.

The service imports that data into storage and later allows users or other systems to export it in different formats.

You are free to choose the technologies, libraries, storage solution and architecture you consider appropriate. Explain your choices where appropriate.

The data

The external system sends CSV files. A batch may look like this:

customers_01.csv

```
id,name,email,age,country
1,John Smith,john@example.com,35,Portugal
2,Jane Doe,jane@example.com,28,Spain
3,Bob Smith,bob@example.com,42,France
```

customers_02.csv

```
id,name,email,age,country,phone
1,John Smith,john@example.com,35,Portugal,+351910000000
4,Ana Costa,ana@example.com,,Portugal
```

customers_03.csv

```
id,name,phone,email,age,country
1,John Smith,+351910000000,john@example.com,35,Portugal
5,Marco Rossi,+390000000000,marco@example,thirty,Italy
```

Requirements

1 Import

- Provide a way to import a CSV file into storage.
- The imported information must be persisted.
- The application should provide enough information to determine whether an import was successful and, where applicable, which records could not be processed.

2 Input format

- Files are produced by an external system and must be processed according to the information they contain.
- Define the behaviour for fields that are missing, unknown, or contain invalid values.

3 Multiple imports

- The external system may send multiple files containing related data.
- Some information may appear in more than one file.
- The application must define how imported information is handled when the same data is received more than once.

4 Storage

- Choose the storage technology you consider appropriate for the problem.
- The application must be able to persist and retrieve the imported information.

5 Export

Stored information must be exportable as **CSV**, **TXT** and **XLS/XLSX**.

The caller specifies which columns to include and in which order:

```
{ "format": "CSV", "columns": ["id", "name", "email", "country"] }
```

```
{ "format": "TXT", "columns": ["name", "email"] }
```

The output should contain the requested fields, in the requested order.

6 Interface

Provide a REST API **or** a CLI supporting:

Import	Import a CSV file
Export	Export data · select format · select columns